The empirical recurrence for OEIS A234917: recovering a conjecture that is not written down

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7 September 2026

Abstract

OEIS A234917 records "Empirical recurrence of order 54", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 1280$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 54, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 54 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A234917 is "Number of (n+1) X (4+1) 0..4 arrays with every 2 X 2 subblock having its diagonal sum differing from its antidiagonal sum by 2, with no adjacent elements equal (constant-stress tilted 1 X 1 tilings).". It has offset 1 and begins

$$1200, 3912, 13336, 55492, 221272, 990956, 4151672, 19178920, \ \dots$$

The entry states:

> Empirical recurrence of order 54 (see link above).

As of the "Last modified" line on the live entry (Jun 20 2022 15:39:19, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1280$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 1280.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2560$ exact terms of the model returns order 54 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{54} c_i \, a(n-i),$$

c_1	3	c_{15}	-41388783951	c_{29}	350268005205456	c_{43}	-43918670272352
c_2	135	c_{16}	-102566483121	c_{30}	96228897593083	c_{44}	10078559738016
c_3	-398	c_{17}	302500366692	c_{31}	-526993876155387	c_{45}	10965684611840
c_4	-8195	c_{18}	570363618575	c_{32}	-82736358928806	c_{46}	-2864095027136
c_5	23819	c_{19}	-1727742951351	c_{33}	636437305698051	c_{47}	-1894296401088
c_6	296441	c_{20}	-2479561179760	c_{34}	37999007896786	c_{48}	543546480448
c_7	-852598	c_{21}	7797453655671	c_{35}	-613461359194141	c_{49}	209568989952
c_8	-7146389	c_{22}	8472683311871	c_{36}	12923804329383	c_{50}	-64429294848
c_9	20425393	c_{23}	-28040292188104	c_{37}	467880882609304	c_{51}	-12890907648
c_{10}	121807013	c_{24}	-22784822084459	c_{38}	-41318250352048	c_{52}	4164590592
c_{11}	-347658826	c_{25}	80826845814517	c_{39}	-278901762663592	c_{53}	313528320
c_{12}	-1522026604	c_{26}	48027198881331	c_{40}	40252187981368	c_{54}	-104509440
c_{13}	4362418630	c_{27}	-187446843791364	c_{41}	127746528473152		
c_{14}	14288112349	c_{28}	-78375303416072	c_{42}	-24387804180512		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 54, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $54 + 4$ terms removed returns order 54 as well, so $\deg q_{\min} = 54 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. ☐

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $54+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 54 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A234917.

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